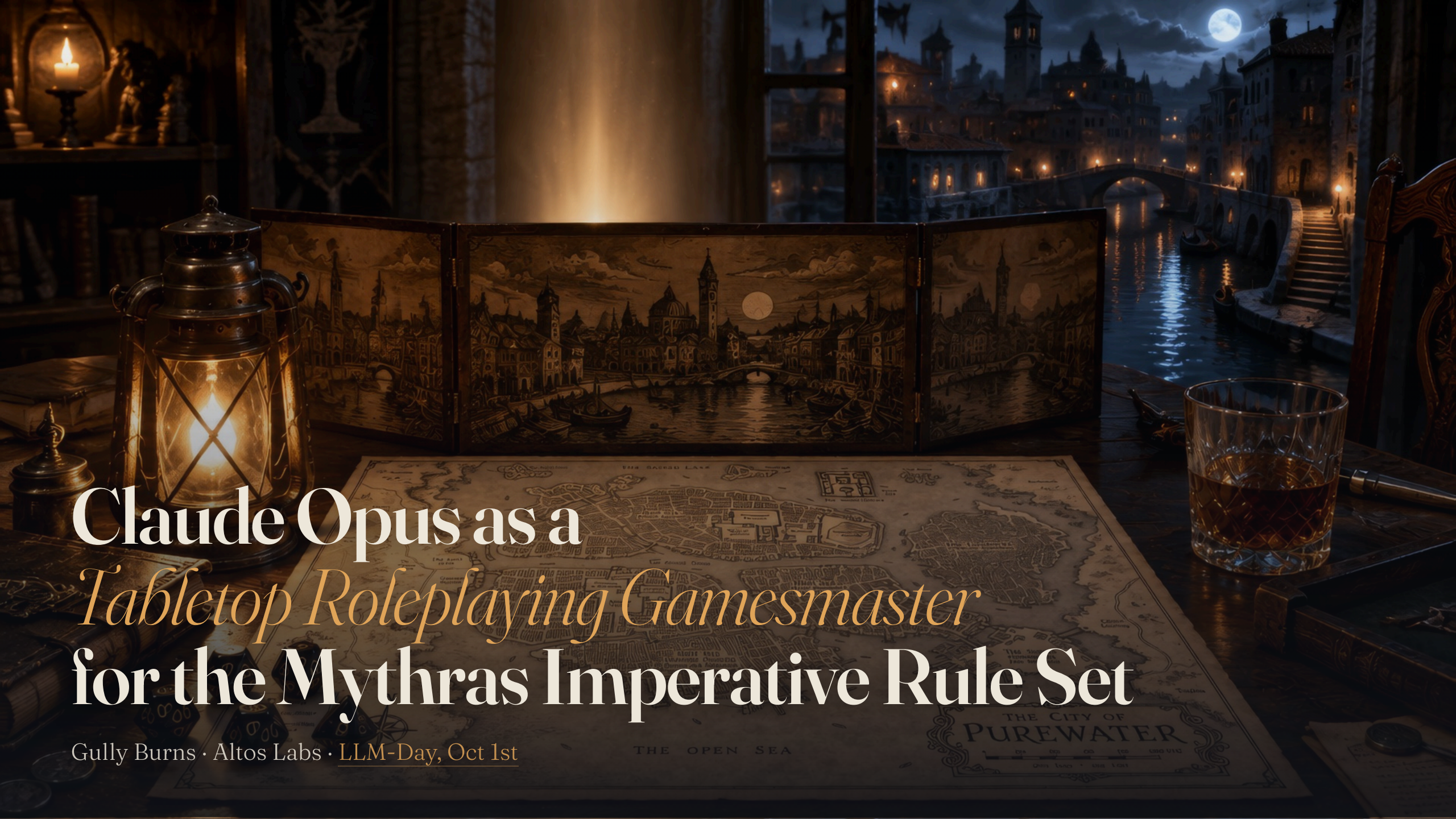




Claude Opus as a *Tabletop Roleplaying Gamesmaster* for the Mythras Imperative Rule Set

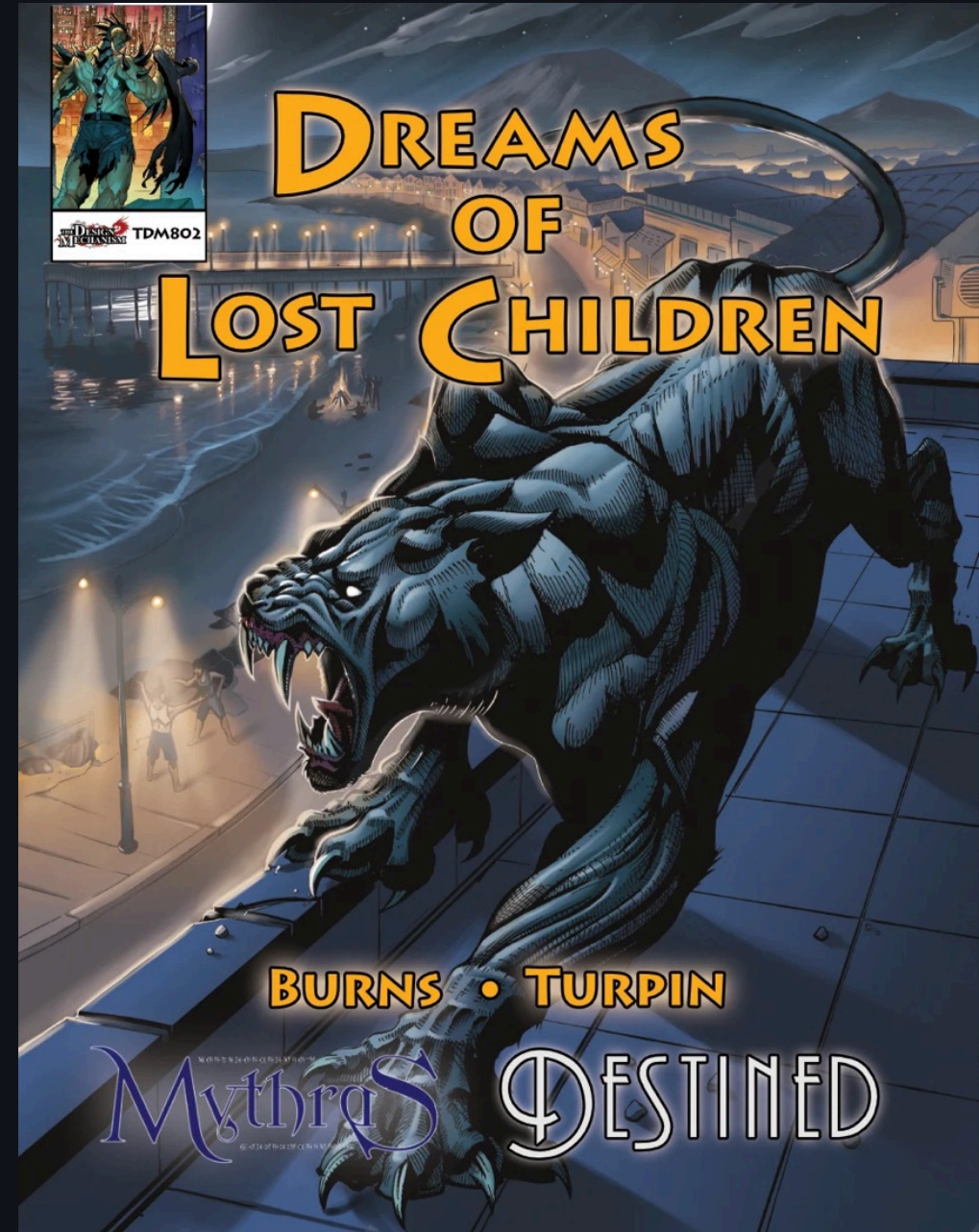
Gully Burns · Altos Labs · [LLM-Day, Oct 1st](#)

Disclaimer

None of the work in this talk involved Altos Labs in any way. It was done on my own time, on my own hardware, with my own accounts. No Altos resources, data, code, or people were used, and nothing here represents Altos Labs' views, products, or plans.

It is a hobby project about running games, built by someone who wanted to play one.

- **Professionally**, I work in the field of scientific knowledge-engineering. Recent work is concerned with '[Agentic Ontological Notebook Memory](#)', forming the basic agent memory framework this talk is built on.
- **I have been playing tabletop roleplaying games since I was 10.** I develop open source systems and material here: [fourth-wall-gaming](#)
- **I am a published author.** The Design Mechanism just released [Dreams of Lost Children](#) (Aug. 25, 2026), a scenario for *Destined*, a superhero game based on the Mythras system.





Stories *transport* us. A role-playing
game transports us as a *participant*.

A language model can now be the narrator.

'Narrative Transportation'

1. **Narrative Transportation is a psychological measure of how audience members engaged in a story are transported by it.** - it is a 15-item questionnaire. Green & Brock 2000
2. **Narrative Transportation is increased by the primary features of role-playing game:** interactivity, user control, perceived contingency, perspective-taking, imagination, mental simulation, spatial presence. Thomas & Grigsby 2024, Table 2
3. **The field has predicted this medium and not measured it** — co-created stories where "the consumer will be the character", via "story prompts provided to AI interfaces". No LLM study among the 95 reviewed. Thomas & Grigsby 2024, §5.3
4. **Evidence is thin.** AI short stories transported readers slightly less than student-written ones ($d = 0.21$); a plain ChatGPT dungeon master scored 2.67/5 on immersion; a GM with memory and world state beat a prompt-only GM on immersion ($N = 12$). Appel et al. 2025; Triyason 2023; Jørgensen et al. 2025
5. **Let's be Careful.** Developing highly compelling systems for storytelling could cause harm to some psychologically vulnerable people.

Green & Brock (2000) *J. Personality & Social Psychology* 79(5):701–721 · van Laer, de Ruyter, Visconti & Wetzels (2014) *J. Consumer Research* 40(5):797–817 · Thomas & Grigsby (2024) *Psychology & Marketing* 41:1805–1819 · Appel, Malecki, Messingschlager & Winkler (2025) "I, ChatGPT" *Humanities & Social Sciences Communications* 12:1892 · Triyason (2023) "Exploring the Potential of ChatGPT as a Dungeon Master" *IAIT 2023*, doi:10.1145/3628454.3628457 · Jørgensen, Tharmabalan, Aslan, Hansen & Merritt (2025) "Static vs. Agentic Game Master AI" arXiv:2502.19519

Steven Johnson Turned Storybooks into Games

- Johnson, an author and editorial lead on Google's NotebookLM, noticed users using a million-token context window as a **RPG campaign notebook**
- Johnson wrote prompts that turn *any source document* into a playable simulation on Gemini 1.5 Pro.
- Three experiments:
 - **The Ghost Map** as a cholera-detective simulation in 1854 London
 - **The Cuban Missile Crisis**, sourced from Wikipedia
 - **One chapter of The Infernal Machine** as a published fingerprint-analysis mini-game

"One interesting application that we noticed early on in the NotebookLM Discord was that long context was enabling role-playing-game enthusiasts to keep track of their games."

Steven Johnson, In the Context of Long Context — Adjacent Possible



Mythras: a complete, mature,
realistic, and *risky* RPG system.

What Mythras is

- **Mature.** A system from The Design Mechanism (Nash and Whitaker), published as RuneQuest 6 in 2012 and as Mythras since 2016 — with a lineage that originates with the original RuneQuest of 1978.
- **Complete across genres.** One core engine runs fantasy, semi-historical worlds (*Mythic Babylon*, *Mythic Rome*, *Mythic Britain* in the Dark Ages, *Mythic Constantinople* in the fifteenth century) — Jack Vance's *Lyonesse*, Bryan Talbot's *Luther Arkwright*, superheroes (*Destined*), science fiction (*M-Space*), with community-contributed genres extending the universes of possible play.
- **Realistic.** Well-developed mechanisms: skills as percentages, opposed and differential rolls, action points, combat styles, hit locations with armor per location, special effects on a won exchange, fatigue, passions that push a roll, several frameworks for magic, superpowers, etc.
- **Risky.** Combat is dangerous. A single won exchange can cripple or kill outright. It is easy to die in Mythras.

The Design Mechanism · *Mythras Imperative* and *Classic Fantasy Imperative* are free under the ORC licence.

How things happen in Mythras

- Named 'skills' manage capabilities: combat, brawn, stealth, perception, influence, deceit, acrobatics, etc..
- A simple action requires a d100 roll under your skill's score.
- A contested action is two rolls. Both sides roll against their own skill and the higher roll that still succeeded wins.
- Game mechanics exist to permit the Gamesmaster (GM) to adjudicate most situations.
- **The point.** Every situation is resolved faithfully by the dice. It is a probabilistic model built from a small number of simple rules, whose outcomes compound — and nobody, the GM included, gets to choose them.

A contest, adjudicated — Magda attempts to watch a stranger without being seen: Stealth vs. Perception

Magda (Stealth 68) rolls 39 → success vs the young man (Perception 45) rolls 52 → failure

She wins outright. He walks under the overhang and never looks up.

...had he rolled 21 → success. Equal grades; her 39 is the higher successful roll.

She still wins, narrowly — he glances at the dark and sees nothing.

...had he rolled 45 → a success greater than Magda's roll.

He stops on the stair and looks straight at her.



A Gamesmaster *manifests*
a world model for players.

What it's like to play

The Fishmarket bridge is a hump of stone over the cut, and at this hour it should be solid with handcarts.

It isn't, because there are two men standing on the crown of it. One telling a story and another listening.

Dragon-blazoned harness, good kit. Not doing anything. One of them has his back to the traffic and is eating something.

The whole morning's trade is going round them in single file, hard against the parapet, and nobody is saying a word about it.

The three calls you've got left are on the other side.

player

I'll walk up, friendly like.

‘Ullo then lads. How are you on this fine morning?

The storyteller stops mid-sentence — because you are eight feet of half-orc coming up a bridge at him out of the fish-market at dawn, and this is the most interesting thing that has happened since Anminster.

The younger one has his hand on his belt near the hilt. It went there fast, and he has not decided to move it away again.

“Morning,” he says, delighted, and he means it. “God's teeth. What do they feed you?”

Behind you the bridge is backing up. Six or seven carts now, and every one of those people is watching you and pretending not to.

player

They feed me fish. Listen, I have work to do on the other side of this bridge and this is just taking a little too long. So if you don't mind I'll hop on through and then I'll be back in a little while. Just thought I'd, y'know, ask nicely.

He laughs — a real one, from the belly — and steps sideways. “Go on then, fish.” He's still grinning. “Nicely asked.”

Six carts have been waiting. Not one of them has been let through. Nobody has told them no. Nobody has told them anything at all.

Halfway across he calls after you: “The Baron's in today. Twenty-five of us over that water before dark, and we're all going to want feeding.”

player

Come back. The Baron? Which one?

“Di Teufel. Baron Hanzo di Teufel, out of the east. We're his.” Then, leaning in: “And we've got a *man* for the sand.”

“Tarrent.” That's the younger one. First word he's said, and the storyteller stops with his mouth open.

player

Smile at the younger one. Make it look a little scary.

“No. It's OK friend. So you're just holding up the traffic till your lord passes? And requisitioning supplies for your men more generally? Like I said, just asking in a friendly manner... I might be able to help with supply issues, who would be a good person to talk to — in your organization, you know, like a quartermaster or something.”

[[Kag · Willpower (Standard 65) → 21 · success]]

[[the younger knight · Willpower (Standard 55) → 18 · success]]

both succeed — Kag takes it on the higher roll, by three

The smile is a real weapon and you've never had to practise it. The younger one gets about a third of it.

He goes very slightly grey around the mouth and does not step back, which for a man his age in front of a colleague is about as well as it could go.

[[Kag · Commerce (65) → 41 · success]]

a soldier's supply problem, framed as a favour

Tarrent laughs. It's the same laugh as before, but it goes on too long and changes shape while it's happening, and by the end he's looking at you with something close to pity. “*Buy* it?”

“Sweetheart, there's twenty-five of us tonight and four hundred camped out east, and we've been on the road nine weeks. What do you think we've been eating?”

There is no supply chain to sell to. No office, no requisition, no name to ask for. You have just offered to sell fish to men who have stopped understanding why anyone would buy it.

“What's your name?” The younger one. Flat, and his hand came off his belt to ask it, which is somehow worse.

player

Interesting.

I'm Kag. Ask around — if you fellas ever bother to actually talk to the people you're stealing from, you'll find out who I am quick as you like.

Nice to meet you all.

Walk off down the other side.

“Stealing” lands harder than the smile did. Tarrent's face does something complicated — and you're already walking.

The bridge lets you through. Of course it does. **There is nobody on it with the authority to stop anybody**, which is the entire discovery of the morning.

Behind you, the younger one says your name once, quietly, to himself. Fixing it.

And then somebody in the queue *laughs*. Just one, cut off fast, and nobody looks up. But it's out now — and by tomorrow she will have been ten feet tall.

Why a language model alone will not do this

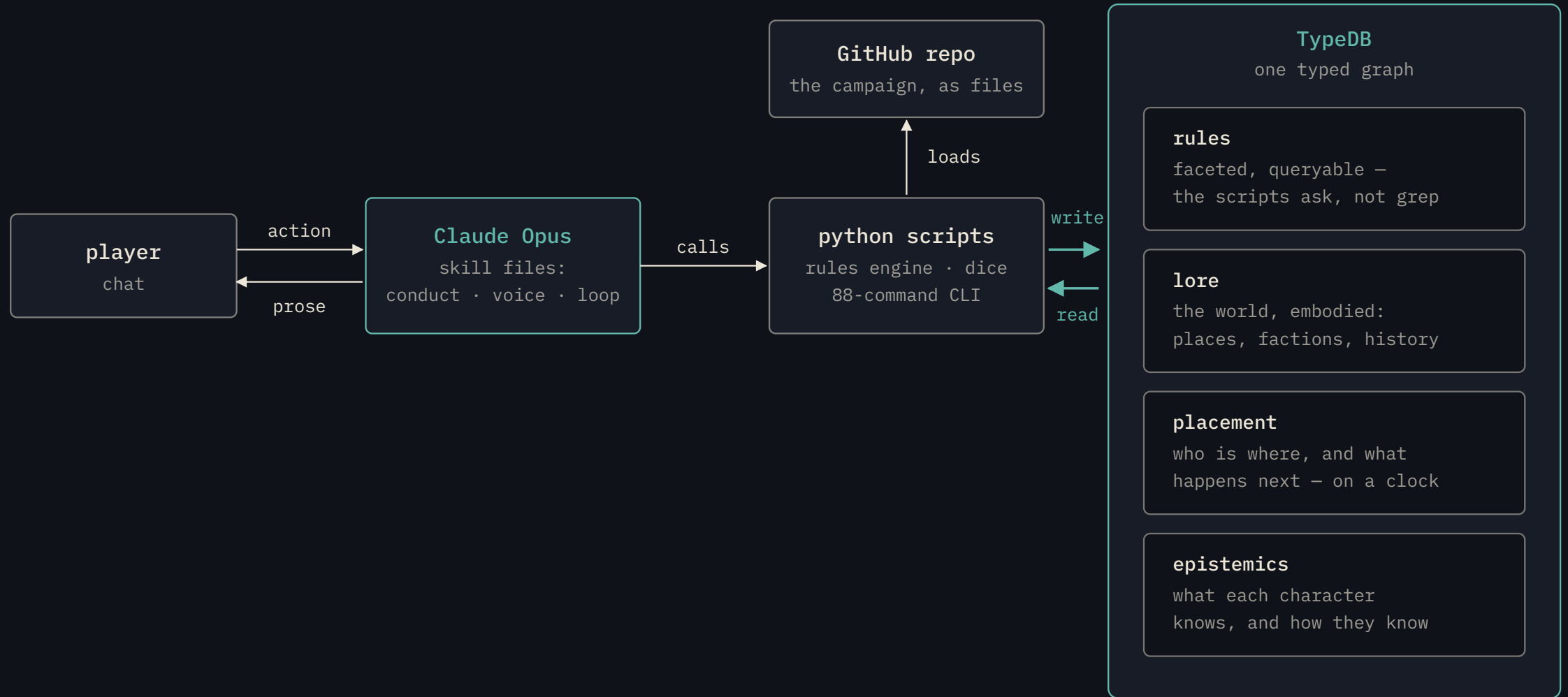
- **It needs a rules engine.** Dice have to decide, and they have to be able to go badly. A narrator who cannot hurt you is not telling a story, it is agreeing with you.
- **It has to show its working.** The roll, the target, the grade — stated in the open. You cannot trust an outcome you cannot check, and trust is the whole game.
- **It needs a knowledge base with real semantics.** Who is where, who knows what, what is still true. Continuity in prose degrades; continuity in a typed graph does not.
- **It needs a story that moves.** Beats on a clock, and characters with their own intentions, so the context the player is acting in changes whether or not they are looking at it.
- **It has to know what to skip.** Left alone it will narrate every step of the walk to the door. Filmmakers call it *shoe leather*; the job is to cut to where something is at risk.
- **What it invents, it must keep.** A name improvised on Tuesday is canon on Friday — so anything made up at the table has to be written down where the next session will find it.

Ontological Notebook Memory



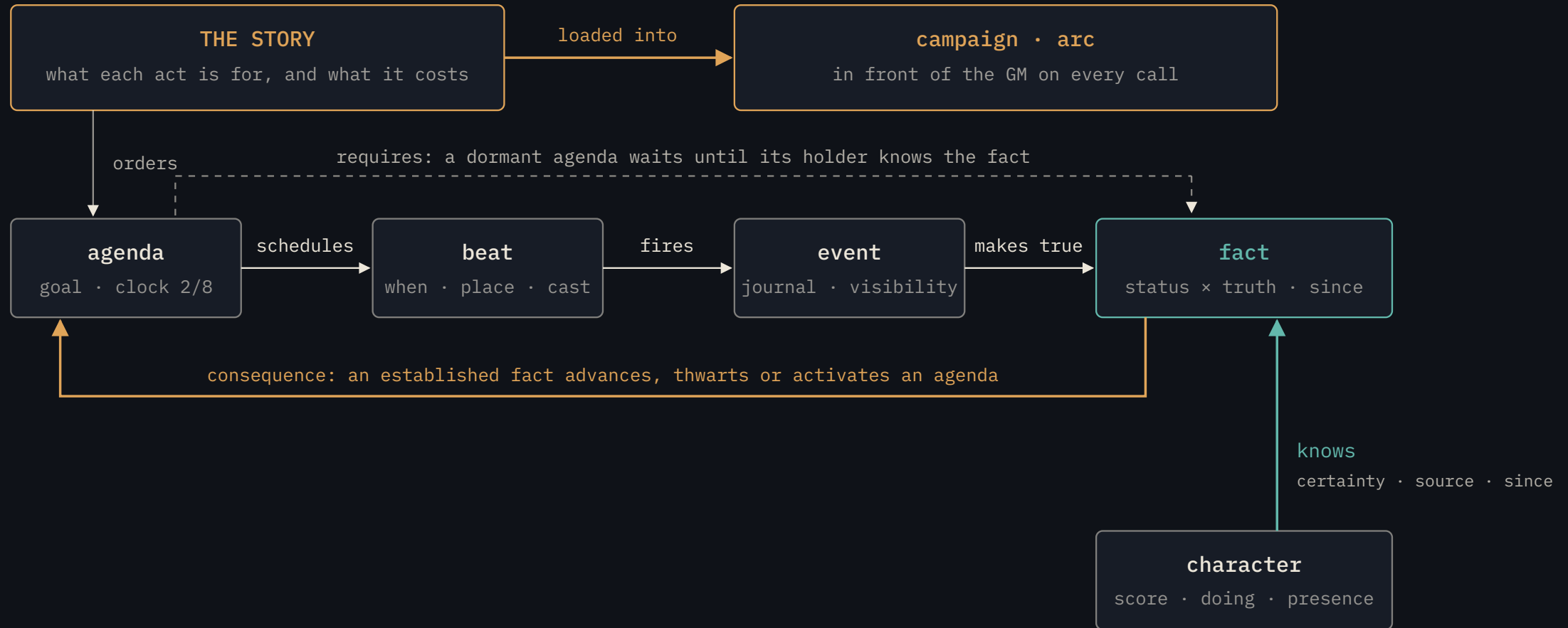
An agent skill ships its own **schema**, and loading it extends what the agent can represent — and remember. Every agent that keeps memory commits to a data model (even if that model is just structured prompt text + responses). Mythras-GM is based on a game-oriented data model - which I will now describe.

Basic System Architecture

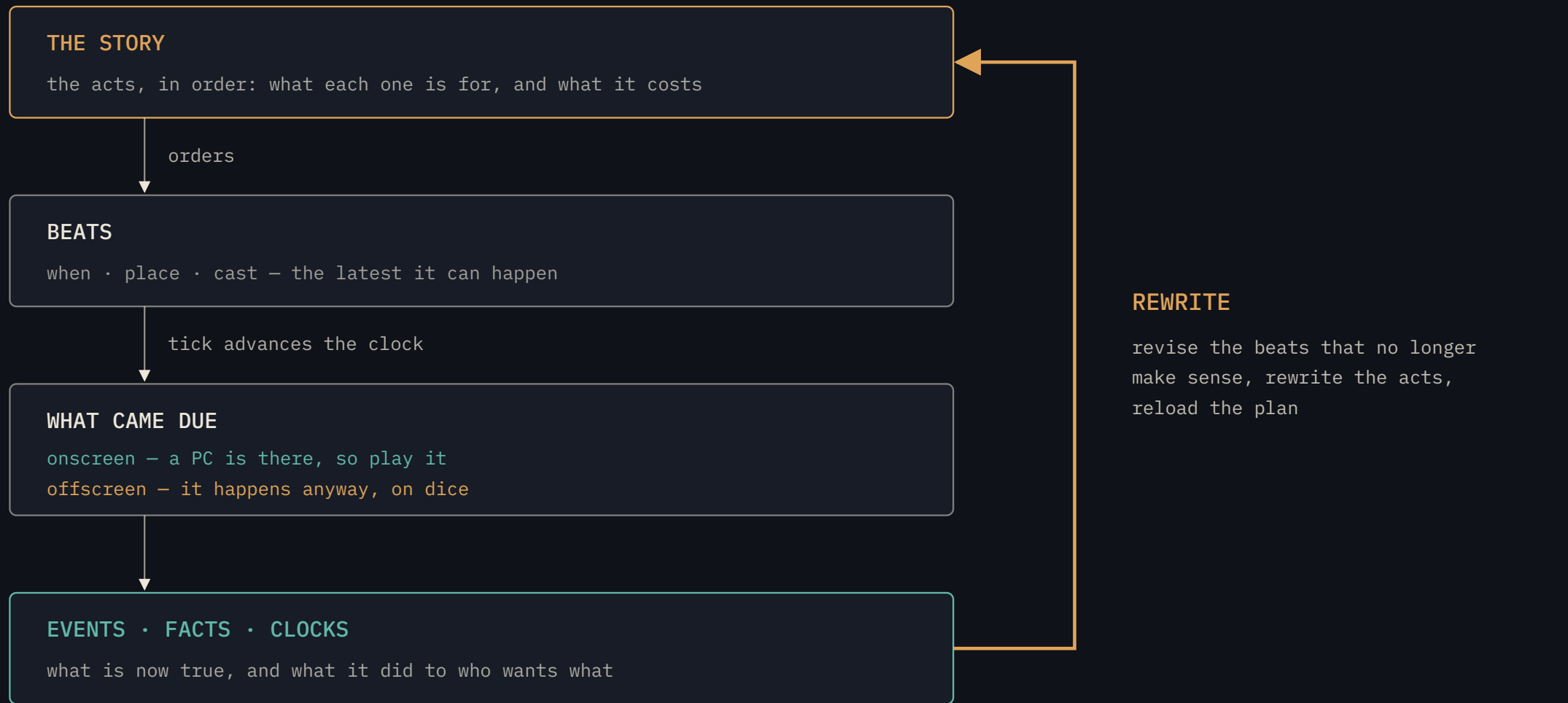


High-level data architecture of the game system, based only on agentic skills with a TypeDB structured memory store.

An Ontological Framework for Stories

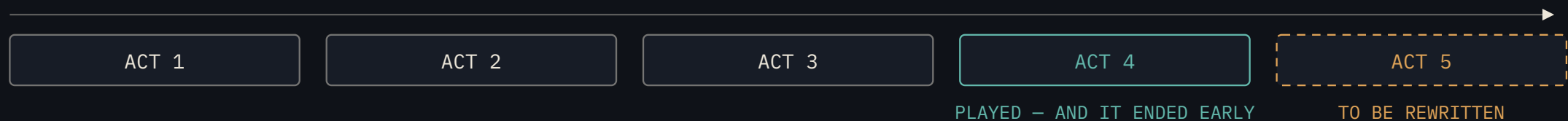


34 entity and relation types. The plan is a thin file of acts — no scene descriptions, those live in the beats — parsed into the save so it cannot decay out of context. Situational truth lives in one place, the fact graph, and what a character knows is a projection through *knows* edges, so two views cannot drift apart.

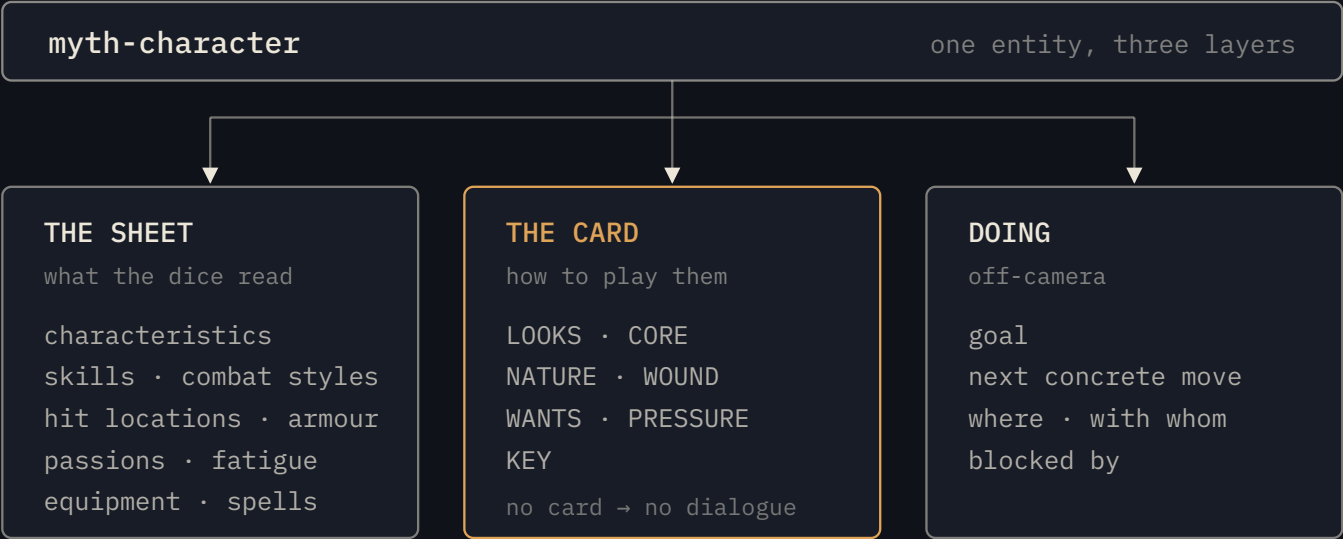


The plan orders the beats, the beats meet the dice, the dice make facts — and the plan is then rewritten to describe the game actually being played.

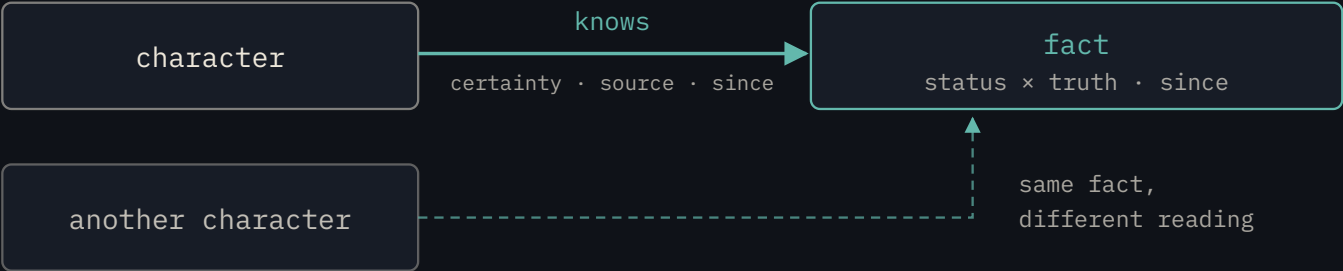
world clock



Characters: Stats, Personality, & Knowledge



knowledge is not on the character – it is an edge



certainty
knows · believes · suspects · wrong

source
witnessed · told · deduced · rumour

status and truth are independent, so a sincerely believed falsehood is data.

a debt is a belief plus a feeling – there is no ledger of obligations anywhere.

one graph, so two views cannot drift: check-consistency catches a character who knew it before it was true.

The sheet is what the dice read, the card is what the actor reads, and *doing* is what they are occupied with when the party is elsewhere. What they **know** is deliberately none of those — it is an edge into the single fact graph, so nobody can act on what they were never in the room for.



Future *Gamesmastering* with LLMs.

How Could This Work Impact Tabletop RPGs

1. **This must make the hobby better.** Many gamers see LLMs as slop that is hollowing out their craft. The only version of this worth building is one that earns its place in that community by (A) making games better, and (B) bringing more people to the hobby.
2. **Fully populated worlds.** The knowledge base will hold as many characters and places as fit on disk. A continuous world, not a staged one so that games play out anywhere in it.
3. **Text is the cheapest (and, for some, most compelling) world-building medium there is.** No art pipeline, no level design, no dialogue trees, no voice cast. Cheap with the focus on story.
4. **LLM-based GMs solve accessibility.** A frustrating aspect of TTRPGs is finding and scheduling a group - LLM-based Gamesmasters are always available.

Mythras-GM @ Fourth Wall Gaming

The Mythras-GM system is a Claude-Code Plugin:

```
# the campaign brings the engine with it
/plugin marketplace add fourth-wall-gaming/mythras-gm
/plugin install purewater@fourth-wall-gaming
/purewater:start
```

Novelization

A played campaign can be read back out as a book. The journal and player-visible canon come out of the graph, chapters are cut where the *story* turns rather than where the evening stopped, and a style card pins the voice — Hemingway, Moorcock, or mine (Gully).

pandoc and Typst typeset it; Claude generates prompts for illustrations.

- **Written the way it was played.** Dialogue comes from the play transcript, never the journal summary — a summary carries what happened and almost none of what was said.
- **Canon flows one way.** Nothing written in the book goes back into game state, so you can keep as many parallel novelizations as you like.

And Then the Dragons Came. A fantasy canal city hosting a tourney welcomes a Baron, whose soldiers carry Dragon regalia. Dark magic and a plot to take over the kingdom ensue.



The Veilwrack: The Stilling. A sky realm with no ground and winged characters who fly. The wind is dying, and some are trying to hasten it on its way.





Thank you.

<https://fourth-wall-gaming.github.io/> · <https://gullys.website>